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Arthur Cayley

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405. An Eighth Memoir on Quantics

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Table No. 82.

Degree, Order, Number, Constituents, quoad the quintic itself A, the quadric covariants B, C, cubic covariants D, E, F, the quartic covariants G, H, I, the quintic covariants J, K, L, and the sextic covariants M, N of § 143, N = new covariant, S = syzygy.

Degree	Order	No.	Constituents
			viz. the absolute constant unity. K
1	5		A
2	10		A^2
	6		C
			B
	2		
3	15		A^3
			AG
	11		F
	9		AB
	7		E N.
	5		D N.
4	20		A^4
	16		A^2C
			AF
	12		C^2, A^2B
	10		AE
	8		AC, EG
	6		I N.
	4		H, B^2 N.
			G N.
5	25		
	21		A^3G
	19		A^2F
			AC
	17	2	A^2B, $\ldots$
	15	2	A^2E, GF
	13	2	A^2D, ABC
	11	2	$AI + BF - CE = 0$ S.
		3	AB^2, AH, CD
	7	2	BE, L F.
	5	2	AG, BD
			K
	3	1	M
	1	1	J N.
6	30	1	
	26	1	A^4C
	24	1	A^3F
	22	2	A^2C^2, A^3B, $\ldots$
	20	2	A^2E, $A^2\ldots$
	18	3	$E^2 + 4C^3 + A^2D - A^2BG = 0$ S.
	16	2	$A(AI + BE - CE) = 0$ $\dot{S}$.
	14	4	$-6AGD - \frac{1}{4}EF - \frac{1}{4}BC^2 + A^2H = 0$, A^2B^2 S.
	12	3	$AL + 3DF - \frac{1}{4}GI = 0$, ABE S.
	10	4	$AB^2G + 4ABD - A^2G + EF \ldots = 0$, GH S.
	8	2	$AK + \frac{1}{4}BI - \frac{1}{2}DE = 0$ S.
	6	4	$AJ + \frac{1}{4}BH - \frac{1}{2}E^2 - CG \ldots = 0$ S.
			N
	4	1	M N.
	2	1	N.

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Article 253 (253)**.** For the explanation of this I remark that the Table No. 81 shows that we have for the degree and order one covariant; this is the absolute constant unity; for the degree 1 and order 5, 1 covariant, this is the quintic itself, A; for degree 2 and order 10, 1 covariant; this is the square of the quintic, A^2; for same degree and order 6, 1 covariant, which had accordingly to be calculated, viz. this is the covariant C; and similarly whenever the Table No. 81 indicates the existence of a covariant of any degree and order, and there does not exist a product of the covariants previously calculated, having the proper degree and order, then in each such case (shown in the last preceding Table by the letter N) a new covariant had to be calculated. On coming to degree 5, order 11, it appears that the number of asyzygetic invariants is only $= 2$, whereas there exist of the right degree and order the 3 combinations AI, BF, CE; there is here a syzygy, or linear relation, between the combinations in question; which syzygy had to be calculated, and was found to be as shown, $AI + BF - CE = 0$, a result given in the Second Memoir, p. 126. Any such case is indicated by the letter S. At the place degree 6, order 16, we find a syzygy between the combinations A^2I, $A \cdot BF$, $A \cdot CE$; as each term contains the factor A, this is only the last-mentioned syzygy multiplied by A, not a new syzygy, and I have written $\dot{S}$ instead of S. The places degree 6, orders 18, 14, 12, 10, 8, 6 indicate each of them a syzygy, which syzygies, as being of the degree 6, were not given in the Second Memoir, and they were first calculated for the present Memoir. It is to be noticed that in some cases the combinations which might have entered into the syzygy do not all of them do so; thus degree 6, order 14, the syzygy is between the four combinations ACD, EF, BG^2, A^2H, and does not contain the remaining combination A^2B^2. The places degree 6, orders 4, 2, indicate each of them a new covariant, and these, as being of the degree 6, were not given in the Second Memoir, but had to be calculated for the present Memoir.

Article 254 (254)**.** I notice the following results:

$$\begin{array}{rc}
\text{Quadrinvt.} & 6H = 3G^2, \\
\text{Cubinvt.} & 6H = -GP + 54GQ, \\
\text{Disct.} & (a \cdot B + \tfrac{1}{4}M) = (-G, Q, -H \frown a, \tfrac{1}{4}\xi)^2, \\
\text{Jac.} & (B, H) = 6J, \\
\text{Hess.} & \dfrac{M}{D} = N,
\end{array}$$

the last two of which indicate the formation of the covariants given in the new Tables $M =$ No. 83 and $N =$ No. 84: viz. if to avoid fractions we take 3 times the covariant D, being a cubic $(a, \ldots \mid x, y)^3$, then the Hessian thereof is a covariant $(a, \ldots)^6(x, y)^6$, which is given in Table, No. 83; and in like manner if we form the Jacobian of the Tables B and H which are respectively of the forms $(a, \ldots)^2(x, y)^2$, and $(a, \ldots)^4(x, y)^4$, this is a covariant $(a, \ldots)^6(x, y)^6$, and dividing it by 6 to obtain the coefficients in their lowest terms, we have the new Table, No. 84. I have in these, for greater distinctness, written the numerical coefficients after instead of before, the literal terms to which they belong.

The two new Tables are:

Table No. 83. $M = (a, \ldots)^6(x, y)^6$. See § 143.

Table No. 84. $N = (a, \ldots)^6(x, y)^6$. See § 143.

Article No. 255. Formulae for the canonical form $ax^5 + by^5 + cz^5 = 0$, where $x + y + z = 0$.

Article 255 (255)**.** The quintic $(a, b, c, d, e, f \mid x, y)^5$ may be expressed in the form

$$au^5 + bv^5 + cw^5,$$

where u, v, w are linear functions of (x, y) such that $u + v + w = 0$. Or, what is the same thing, the quintic may be represented in the canonical form

$$ax^5 + by^5 + cz^5,$$

where $x + y + z = 0$; this is $= (a - c, -c, -c, -c, -c, b - c \mid x, y)^5$, and the different covariants and invariants of the quintic may hence be expressed in terms of these coefficients (a, b, c).

For the invariants we have

$$\begin{aligned} G = J &= b^2c^2 + c^2a^2 + a^2b^2 - 2abc(a + b + c), \\ Q = K &= a^2b^2c^2(bc + ca + ab), \\ -\Pi = L &= a^4b^4c^4, \\ W = I &= 16a^3b^3c^3(b - c)(c - a)(a - b). \end{aligned}$$

[Observe that throughout the present Memoir, the invariants, instead of being called G, Q, $-\Pi$, W are called I, J, K, L, viz. the I, J, K, L in all that follows denote the invariants, and not the covariants denoted by these letters in § 142, 143. Moreover D is used to denote the invariant Q^2, which is in fact the discriminant of the quintic.]

Hence, writing for a moment

$$a + b + c = p, \quad bc + ca + ab = q, \quad abc = r,$$

and therefore

$$J = q^2 - 4pr, \quad K = r^2q, \quad L = r^4,$$

we have

$$(a - b)^2(b - c)^2(c - a)^2 = p^2q^2 - 4q^3 - 4p^3r + 18pqr - 27r^2,$$

and thence

$$I^2 = 16q^3r^6(p^2q^2 - 4q^3 - 4p^3r + 18pqr - 27r^2),$$

and

$$\begin{aligned} &J(K^2 - JL)^2 + 8K^3L - 72JKL^2 - 432L^3 \\ &= r^{12}\{(q^2 - 4pr)\,16p^2 + 8q^6 - (q^2 - 4pr)\,72q^3 - 432r^3\} \\ &= 8r^{12}\{(q^2 - 4pr)(2p^2 - 9q) + q^3 - 54r^2\} \\ &= 16r^{12}[p^2q^2 - 4q^3 - 4p^3r + 18pqr - 27r^2], \end{aligned}$$

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Article 256 (255 *continued*)**.** that is,

$$I^2 = J(K^2 - JL)^2 + 8K^3L - 72JKL^2 - 432L^3,$$

which is the simplest mode of obtaining the expression for the square of the 18-thic or skew invariant I in terms of the invariants J, K, L of the degrees 4, 8, 12 respectively.

If instead of the invariant K of the degree 8 we consider the invariant D [$= Q^2$ as before-mentioned] of the same degree, this is

$$\begin{aligned} D &= [b^2c^2 + c^2a^2 + a^2b^2 - 2abc(a+b+c)]^2 - 128a^2b^2c^2(bc+ca+ab), \\ &= q^4 - 8q^2pr - 128q^3r^2 + 16p^2r^2, \\ D &= \text{Norm}\,((bc)^2 + (ca)^2 + (ab)^2) \end{aligned}$$

and we have also the following covariants

$$\begin{aligned} B &= (-ac, ab - ac - bc, -bc \mid x, y)^2 = bc\,yz + ca\,zx + ab\,xy. \\ C &= (-ac, -3ac, -3ac, ab - ac - bc, -3bc, -3bc, -bc \mid x, y)^4 \\ &= bc\,y^2z^2 + ca\,z^2x^2 + ab\,x^2y^2. \\ D &= (0, -abc, -abc, 0 \mid x, y)^3 = abc\,xyz. \\ E &= (ab^2 - ac^2 + bc^2 - a^2c - 2abc)\,x^5 \\ &\quad + (-5ac^2 + 5bc^2 - 5abc)\,x^4y \\ &\quad + (-10ac^2 + 10bc^2 - 2abc)\,x^3y^2 \\ &\quad + (-10ac^2 + 10bc^2 + 2abc)\,x^2y^3 \\ &\quad + (-5ac^2 + 5a^2bc)\,xy^4 \\ &\quad + (-ab^2 - ac^2 + bc^2 + b^2c + 2abc)\,y^5 \\ &= (b-c)\,a^2x^5 + (c-a)\,b^2y^5 + (a-b)\,c^2z^5 \\ &\quad - abc(y-z)(z-x)(x-y)(yz+zx+xy). \end{aligned}$$

Article No. 256. Expression of the 18-thic Invariant in terms of the roots.

Article 257 (256)**.** It was remarked by Dr Salmon, that for a quintic $(a, b, c, d, e, f \mid x, y)^5$ which is linearly transformable into the form $(a, 0, c, 0, e, 0 \mid x, y)^5$, the invariant I is $= 0$. Now putting for convenience $y = 1$, and considering for a moment the equation

$$x(x-\beta)(x-\gamma)(x-\delta)(x-\epsilon) = 0,$$

then writing $\dfrac{1 - mx}{1 - m\alpha}$ herein for x, the transformed equation is

$$X(X-\beta')(X-\gamma')(X-\delta')(X-\epsilon') = 0,$$

where

$$\beta' = \frac{\beta - \alpha}{1 - m\beta}, \quad \gamma' = \frac{\gamma - \alpha}{1 - m\gamma}, \quad \delta' = \frac{\delta - \alpha}{1 - m\delta}, \quad \epsilon' = \frac{\epsilon - \alpha}{1 - m\epsilon};$$

hence m may be so determined that $\delta' + \epsilon'$ may be $= 0$; viz. this will be the case if $\delta + \epsilon = 2m\delta\epsilon$, or $m = \dfrac{\delta + \epsilon}{2\delta\epsilon}$. In order that $\beta' + \gamma'$ may be $= 0$, we must of course have $m = \dfrac{\beta + \gamma}{2\beta\gamma}$, and hence the condition that simultaneously $\beta' + \gamma' = 0$ and $\delta' + \epsilon' = 0$ is

$$\frac{\beta + \gamma}{2\beta\gamma} = \frac{\delta + \epsilon}{2\delta\epsilon};$$

that is, $(\beta + \gamma)\delta\epsilon - \beta\gamma(\delta + \epsilon) = 0$. Or putting $x - \alpha$ for x and $\beta - \alpha$, $\gamma - \alpha$, $\delta - \alpha$, &c. for β, γ, &c., we have the equation

$$(x-\alpha)(x-\beta)(x-\gamma)(x-\delta)(x-\epsilon) = 0,$$

which is by the transformation $x-\alpha$ into $m(\gamma-\alpha)\dfrac{x-\alpha}{1-m(x-\alpha)}$ changed into

$$(x-\alpha')(x-\beta')(x-\gamma')(x-\delta')(x-\epsilon')=0$$

(where $\alpha'=\alpha$), and the condition in order that in the new equation it may be possible to have simultaneously $\beta'+\gamma'-2\alpha'=0$, $\delta'+\epsilon'-2\alpha'=0$, is

$$(\beta+\gamma-2\alpha)(\delta-\alpha)(\epsilon-\alpha)-(\delta+\epsilon-2\alpha)(\beta-\alpha)(\gamma-\alpha)=0,$$

or, as this may be written,

$$\begin{vmatrix} 1 & 2\alpha & \alpha^2 \\ 1 & \beta+\gamma & \beta\gamma \\ 1 & \delta+\epsilon & \delta\epsilon \end{vmatrix}=0.$$

Article 258 (256 *continued*). Hence writing $x+\alpha'$ for x, the last-mentioned equation is the condition in order that the equation

$$(x-\alpha)(x-\beta)(x-\gamma)(x-\delta)(x-\epsilon)=0$$

may be transformable into

$$X(X-\beta')(X-\gamma')(X-\delta')(X-\epsilon')=0,$$

where $\beta'+\gamma'=0$, $\delta'+\epsilon'=0$, that is, into the form $X(X^2-\beta'^2)(X^2-\delta'^2)=0$. Or replacing x, if we have

$$(a,b,c,d,e,f \mid x,y)^5 = a(x-\alpha y)(x-\beta y)(x-\gamma y)(x-\delta y)(x-\epsilon y),$$

then the equation in question is the condition in order that this may be transformable into the form $(a',0,c',0,e',0 \mid x,y)^5$; that is, in order that the 18-thic invariant I may vanish. Hence observing that there are 15 determinants of the form in question, and that any root, for instance α, enters as α^2 in 3 of them and in the simple power α in the remaining 12, we see that the product

$$\begin{vmatrix} 1 & 2\alpha & \alpha^2 \\ 1 & \beta+\gamma & \beta\gamma \\ 1 & \delta+\epsilon & \delta\epsilon \end{vmatrix}$$

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Article 259 (256 *continued*)**.** contains each root in the power 18, and is consequently a rational and integral function of the coefficients of the degree 18, viz. save as to a numerical factor it is equal to the invariant I. And considering the equation $(a, \dots \mid x, y)^5 = 0$ as representing a range of points, the signification of the equation $I = 0$ is that, the pairs (β, γ) and (δ, ϵ) being properly selected, the fifth point α is a focus in regard to the pairs (β, γ) and (δ, ϵ).

Article No. 257. Expression for I as a Quadratic Function of the Coefficients (a, b, c, d, e, f).

Article 260 (257)**.** Reverting to the formulae

$$\begin{aligned} I &= 16a^3b^3c^3(b-c)(c-a)(a-b), \\ D &= \{b^2c^2 + c^2a^2 + a^2b^2 - 2abc(a+b+c)\}^2 - 128a^2b^2c^2(bc+ca+ab), \end{aligned}$$

if in these we write $b = a + h$, $c = a + k$, then it is found that the resulting expression for I contains the factor a^3, and if we further write $k = lh$, then I contains the factor h^6; hence, attending only to the terms involving a^3h^6, we have

$$\begin{aligned} I &= 16a^3(l-1)l(l+1)(2l-1)(l-2)\,a^3h^6, \\ D &= a^4 + 8a^3h(l^2 - l + 1) + \dots; \end{aligned}$$

and hence the equation $I = 0$ gives $l = \infty$, $l = 1$, $l = 0$, $l = -1$, $l = \frac{1}{2}$, $l = 2$; these correspond respectively to the forms

$$\begin{aligned} &(a, b, b, b, b, b \mid x, y)^5, \quad (a, b, c, c, c, c \mid x, y)^5, \\ &(a, a, a, a, c, c \mid x, y)^5, \quad (a, a, a, c, c, c \mid x, y)^5, \\ &(a, a, a, a, a, c \mid x, y)^5, \quad (a, b, b, b, b, c \mid x, y)^5, \end{aligned}$$

where the first and fifth are not essentially distinct; and these are the several forms for which the skew invariant I vanishes.

Article No. 258. Further investigation of the canonical form.

Article 261 (258)**.** The form $ax^5 + by^5 + cz^5$, where $x + y + z = 0$, has been considered by Sylvester and by Hermite; Sylvester's expression for the quintic is $(\alpha x + \beta y)^5 + (\gamma x + \delta y)^5 - (\alpha x + \beta y + \gamma x + \delta y)^5$, that is, $(\alpha x + \beta y)^5 + (\gamma x + \delta y)^5 + (\epsilon x + \zeta y)^5$, where $\alpha x + \beta y + \gamma x + \delta y + \epsilon x + \zeta y = 0$, which is the same thing as $ax^5 + by^5 + cz^5$, where $x + y + z = 0$. Hermite uses the forms $x^5 + y^5 + z^5$, $x^5 + y^5 - (x+y)^5$, or in general $ax^5 + by^5 + c(x+y)^5$. The transformation from the general form to the canonical form is readily effected by means of the function

$$\begin{vmatrix} a & b & c & d & e \\ b & c & d & e & f \\ x & y & 0 & 0 & 0 \\ 0 & x & y & 0 & 0 \\ 0 & 0 & x & y & 0 \end{vmatrix},$$

Article 262 (258 *continued*)**.** which, equated to zero, gives the covariant C of the Second Memoir; writing this under the form $xy(ax^2+2hxy+by^2)=0$, we have $x=0$, $y=0$, $ax^2+2hxy+by^2=0$; and if in the general quintic we write in place of (x,y) either of the linear factors of $ax^2+2hxy+by^2$, the quintic will be reduced to the form $(a',b',c',d',0,0 \mid x,y)^5$, which can be further reduced to the canonical form. This implies that the quartic function $ax^2+2hxy+by^2$ can be expressed as a product of linear factors, that is, its discriminant must be $=0$, or the function must have a square factor. But the discriminant of the function is the D of the memoir; hence the condition in order that the general quintic may be reducible to the canonical form is $D=0$. But when $D=0$, the covariant C is a perfect square, say $=(lx+my)^4$, and the reduction is effected by writing either $lx+my=0$ or (what is the same thing) by reversing the order of (x,y) and writing $lx+my=0$.

Article No. 259. The Catalecticant.

Article 263 (259)**.** The condition that the quintic $(a,b,c,d,e,f \mid x,y)^5$ may be expressed in the canonical form $(a,0,c,0,e,0 \mid x,y)^5$, is that the catalecticant may vanish; this is

$$\begin{vmatrix} a & b & c & d \\ b & c & d & e \\ c & d & e & f \end{vmatrix} = 0,$$

which is the condition that the quintic may be the sum of three fifth powers. For the form $(a,0,c,0,e,0 \mid x,y)^5$, the catalecticant is at once seen to vanish, and conversely if the catalecticant vanish, the form is reducible to $(a,0,c,0,e,0 \mid x,y)^5$. The catalecticant is an invariant of the degree 3, but it is not a new invariant; in fact it is the quadrivariant $=H$, or say it is $=\frac{1}{6}H$.

Article 264 (260)**.** The condition in order that the quintic may be the sum of two fifth powers is that the form may be $(a,0,0,0,e,0 \mid x,y)^5$, that is, ax^5+ey^5; the conditions here are that not only the catalecticant but also the quadrivariant (which for this form is $=0$) shall vanish; that is, the conditions are $H=0$, $I=0$. But these two conditions imply also the vanishing of the cubinvariant, that is, we have $J=0$; and conversely $H=0$, $I=0$, $J=0$ are the conditions in order that the quintic may be the sum of two fifth powers. Observe that the last two conditions $I=0$, $J=0$ imply $H=0$; for if the quadrivariant do not vanish, the quintic can only be the sum of three fifth powers at least, and if in the sum of three fifth powers $ax^5+by^5+cz^5$, we have $c=0$, then the form reduces itself to the sum of two fifth powers.

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Article 265 (261)**.** For the canonical form $ax^5+by^5+cz^5$, $x+y+z=0$, we have I, J, K, L as above; we have also $D=Q^2$, and we can express I^2 as a function of J, K, L; but it is further possible to express I^2 as a function of J, D, L. We have

$$\begin{aligned} D &= \text{Norm}\,\{b^2c^2+c^2a^2+a^2b^2-2abc(a+b+c)\} \\ &= \text{Norm}\,\{(bc)^2+(ca)^2+(ab)^2-2abc\cdot p\} \\ &= q^4-8q^2pr-128q^3r^2+16p^2r^2, \end{aligned}$$

and hence I^2 can be expressed as a function of p, q, r, or (which is the same thing) of J, D, L. The expression is in fact

$$I^2 = 16r^6\{q^2(p^2q^2-4q^3-4p^3r+18pqr-27r^2)\},$$

and we thence obtain the expression in terms of J, D, L.

Article No. 262. The Sextic Covariants M, N.

Article 266 (262)**.** Reverting to the covariants B and H, and to the expressions for them in terms of the canonical form, we have

$$\begin{aligned} B &= bc\,yz+ca\,zx+ab\,xy, \\ C &= bc\,y^2z^2+ca\,z^2x^2+ab\,x^2y^2, \end{aligned}$$

and thence we obtain the remaining covariants; but the expressions become very complicated. I merely notice that for the canonical form, the covariants M and N (the sextic covariants of § 254) acquire very simple forms. In fact, using the formulae there given,

$$\begin{aligned} 3D &= (a,\ldots \mid x,y)^3 = 3abc\,xyz, \\ \text{Hess}(3D) &= M = (a,\ldots)^6(x,y)^6, \\ \text{Jac}(B,H) &= 6N, \end{aligned}$$

we find without difficulty

$$\begin{aligned} M &= -a^2b^2c^2\,x^2y^2z^2, \\ N &= a^3b^3c^3\,xyz(yz+zx+xy), \end{aligned}$$

which are the required expressions; it is to be observed that in the expression for N, the factor xyz belongs to the covariant D of the memoir.

Article No. 263. The Resolvent Sextic.

Article 267 (263)**.** The equation $(a, b, c, d, e, f \mid x, y)^5 = 0$ may be transformed by a linear substitution into the form $(a', 0, c', 0, e', 0 \mid x, y)^5 = 0$, the condition for the possibility of which is that the skew invariant I shall vanish; and the further condition in order that this may be the sum of two fifth powers, that is, that it may be of the form $a'x^5 + e'y^5 = 0$, is that J shall also vanish. We have thus for the determination of the coefficients of the reduced form the two conditions $I = 0$, $J = 0$; and the ratios $a' : c' : e'$ are thereby determined. In fact if we write the reduced equation under the form

$$ax^5 + 10cx^3y^2 + 5exy^4 = 0,$$

then we have

$$\begin{aligned} I &= 16c^2e^2(ae - 4c^2), \\ J &= 16c^2e^2(ae - 4c^2)^2 - 128c^2e^2 \cdot c^2e^2 \\ &= 16c^2e^2\{(ae - 4c^2)^2 - 8c^2e^2\}, \end{aligned}$$

and hence $I = 0$ gives either $c = 0$, or $e = 0$, or $ae - 4c^2 = 0$. If $c = 0$, the equation is $x(ax^4 + 5ey^4) = 0$; if $e = 0$, it is $x^3(ax^2 + 10cy^2) = 0$. But excluding these cases, the condition $I = 0$ gives $ae - 4c^2 = 0$, that is, $e = \dfrac{4c^2}{a}$; and substituting this value, the equation becomes

$$ax^5 + 10cx^3y^2 + \frac{20c^2}{a}xy^4 = 0,$$

or putting $\dfrac{x}{y} = \theta$, this is

$$a^2\theta^5 + 10ac\theta^3 + 20c^2\theta = 0,$$

that is,

$$a\theta\left\{\left(a\theta^2 + 5c\right)^2 - 5c^2\right\} = 0.$$

The form of the resolvent sextic is found to be

$$\theta^6 - \frac{5J}{K}\theta^4 + 5\theta^2 - 1 = 0,$$

and the roots of this equation determine the different canonical forms to which the quintic can be reduced.

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Article No. 264. Tables for the Covariants of the Quintic.

Article 268 (264)**.** The Tables Nos. 83 and 84 give the two sextic covariants M and N which have been referred to. As already mentioned, for greater distinctness the numerical coefficients are written after instead of before the literal terms to which they belong. The literal parts are written in the order of the powers of x and y, and the numerical coefficients are placed to the right of the terms.

Article No. 265. Geometrical representation of the roots of a quintic equation.

Article 269 (265)**.** The geometrical representation of the roots of a quintic equation is obtained by means of the canonical form $ax^5 + by^5 + cz^5 = 0$, where $x + y + z = 0$. If in the plane we take a triangle ABC, and upon the sides thereof measure off distances x, y, z respectively, such that $x + y + z = 0$ (where the signs are taken account of), then if $x : y : z$ be the roots of the quintic equation, the locus of the point determined by these distances is a curve of the fifth order; but the more precise determination requires a further discussion.

Article No. 266. On the Characters of a Quintic Equation.

Article 270 (266)**.** A quintic equation may have the following characters:

1r 5 real roots: character $5r$,
2r 3 real roots and 2 imaginary roots: character $3r + 2i$,
3r 1 real root and 4 imaginary roots: character $r + 4i$.

The discriminant D is positive for the characters $5r$ and $r + 4i$, and negative for the character $3r + 2i$. The absolute invariants J, K, L (or say the invariants of the degrees 4, 8, 12 respectively) serve to determine the character of the equation. The discriminant D is an invariant of the degree 8, expressible in terms of J, K, L.

Article 271 (267)**.** The character of a quintic equation is completely determined by the absolute invariants together with the sign of the skew invariant I. The absolute invariants are functions of J, K, L which are unaltered by any linear transformation of the equation; and the sign of I distinguishes between characters which have the same absolute invariants.

Article 272 (268)**.** The different characters of the roots correspond to different regions in the space of the invariants, separated by the discriminant-surface $D = 0$. The surface $D = 0$ divides the space into regions, and for each region the character of the equation is constant. The passage from one region to another across the surface $D = 0$ corresponds to a change in the character of the equation, arising from the coalescence of two real roots or of two imaginary roots.

Article 273 (269)**.** The discriminant D being a function of the invariants J, K, L, the surface $D=0$ is a surface in the space (J,K,L). This surface has a nodal line (or cuspidal edge) along which the discriminant and its first derived functions all vanish; and the nodal line corresponds to the case of two pairs of equal roots. The surface has also a triple point corresponding to the case of three equal roots; and the triple point lies upon the nodal line.

Article No. 270. The Bicorn Curve.

Article 274 (270)**.** The form of the Bicorn curve, so far as it is material for the discussion, is shown in the Plate, fig. 2, and it thereby appears that it divides the plane into six regions, which may be called P, Q, R, S, T, U. The equation of the Bicorn is

$$(x^2+2xy-y^2)^2+y^3(2x+y)=0,$$

or, as it may also be written,

$$(x^2+2xy-y^2)^2=-y^3(2x+y).$$

The curve has a node-cusp at the origin, and it has also a cusp at infinity on the line $y=0$. The tangent at the node-cusp is $y=0$, and the tangent at the other cusp is the line infinity. The curve has a double tangent at $x+y=0$, touching at the points where $x^2+2xy-y^2=0$.

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Article 275 (271)**.** The Bicorn curve is a curve of the fourth order and of the fourth class; in fact, the order is 4 and the class is also 4. The node-cusp counts as a node, a cusp, an inflexion, and a double tangent; and the cusp at infinity counts as a cusp, an inflexion, and a double tangent. The node-cusp absorbs $6+8+1=15$ inflexions, and the other cusp 8 inflexions; there remains $24-15-8=1$ inflexion, viz. this is the inflexion at infinity, having the line infinity for tangent; there is not, besides the tangent at the node-cusp, any other double tangent of the curve.

Article 276 (272)**.** The Bicorn divides the plane into six regions P, Q, R, S, T, U, and for each of these regions the values of the invariants J, D, and 2^4L-J^2 have determinate signs. We have thus the following Table:

Region	D	J	2^4L-J^2	Character
P	$+$	$-$	$+$	$5r$
Q	$+$	$-$	$-$	$r+4i$
T	$+$	$+$	$+$	$r+4i$
R	$-$	$-$	$\pm$	$3r+2i$
S	$-$	$-$	$+$	$3r+2i$
U	$-$	$+$	$+$	$3r+2i$

so that we have the kingdom $5r$ consisting of the single region P, the kingdom $r+4i$ consisting of the regions Q and T, and the kingdom $3r+2i$ consisting of the regions R, S, and U.

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Article 277 (273)**.** For a given equation if D is $= -$, the character is $3r + 2i$; if $D = +$, $J = +$, the character is $r + 4i$; if $D = +$, $J = -$, then, according as $2^4L - J^2$ is $= +$ or is $= -$ the character is $5r$ or $r + 4i$. But in the last case the distinction between the characters $5r$ and $r + 4i$ may be presented in a more general form, involving a parameter μ, arbitrary between certain limits. In fact drawing upwards from the origin, as in Plate, fig. 3, the lines $x - 2y = 0$ and $x + y = 0$, and between them any line whatever $x + \mu y = 0$, the point (x, y), assumed to lie in the region P or Q, will lie in the one or the other region according as it lies on the one side or the other side of the line in question, viz. in the region P if $x + \mu y$ is $= -$, in the region Q if $x + \mu y$ is $= +$. But we have

$$x + \mu y = \frac{2^4L - J^2 + \mu JD}{\dots},$$

and J being by supposition negative, the sign of $2^4L - J^2 + \mu JD$ is opposite to that of $x + \mu y$. The region is thus P or Q according to the sign of $2^4L - J^2 + \mu JD$; and completing the enunciation, we have, finally, the following criteria for the number of real roots of a given quintic equation, viz.

If $D = -$, the character is $3r + 2i$,

If $D = +$, $J = +$, then it is $r + 4i$,

If $D = +$, $J = -$, then μ being any number at pleasure between the limits $+1$ and -2, both inclusive, if

$2^4L - J^2 + \mu JD = +$ the character is $5r$,

$2^4L - J^2 + \mu JD = -$, „ „ „ $r + 4i$.

Article 278 (274). The characters $5r$ of the region P and $r+4i$ of the regions Q and T may be ascertained by means of the equation $(a,0,c,0,e,0 \mid x,1)^5=0$, that is

$$x(a\xi^4+10c\xi^2+e)=0;$$

there is always the real root $\xi=0$, and the equation will thus have the character $5r$ or $r+4i$ according as the reduced equation $a\xi^4+10c\xi^2+e=0$ has the character $4r$ or $4ri$. It is clear that (a,e) must have the same sign, for otherwise ξ^2 would have two real values, one positive, the other negative, and the character would be $2r+2i$. And (a,e) having the same sign, then the character will be $4r$, if ξ^2 has two real positive values, that is, if $ae-4c^2$ is $=-$, and the sign of c be opposite to that of a and e, or, what is the same thing, if ce be $=-$; but if these two conditions are not satisfied, then the values of ξ^2 will be imaginary, or else real and negative, and in either case the character will be $4ri$.

Article 279 (275). Now, for the equation in question, putting in the Tables $b=d=f=0$, we find for the invariants

$$\begin{aligned}
J &= 16c^2e^2(ae-4c^2),\\
D &= 16c^2e^2\{(ae-4c^2)^2-8c^2e^2\},\\
L &= a^4e^4,
\end{aligned}$$

whence we can calculate the values of J, D, L for any given values of a, c, e. In particular, the equation $a\xi^4+10c\xi^2+e=0$ has the character $4r$ if $ae-4c^2=-$ and $c=-$, that is, if $J=+$ and $c=-$; but we have $D=+$ when $ae-4c^2=+$, or when $(ae-4c^2)^2-8c^2e^2=+$; and the sign of D being given, the character is determined as above.

Article 280 (276)**.** But if $D = +$, $J = -$, then μ being any number at pleasure between the limits $+1$ and -2, both inclusive, if

$$\begin{aligned} 2^4L - J^2 + \mu JD &= + \quad \text{the character is } 5r, \\ 2^4L - J^2 + \mu JD &= -, \quad \text{,, ,, ,, } r + 4i. \end{aligned}$$

Article 281 (277)**.** The characters $5r$ of the region P and $r + 4i$ of the regions Q and T may be ascertained by means of the equation $(a, 0, c, 0, e, 0 \mid x, 1)^5 = 0$, that is

$$x(a\xi^4 + 10c\xi^2 + e) = 0;$$

there is always the real root $\xi = 0$, and the equation will thus have the character $5r$ or $r + 4i$ according as the reduced equation $a\xi^4 + 10c\xi^2 + e = 0$ has the character $4r$ or $4ri$. It is clear that (a, e) must have the same sign, for otherwise ξ^2 would have two real values, one positive, the other negative, and the character would be $2r + 2i$. And (a, e) having the same sign, then the character will be $4r$, if ξ^2 has two real positive values, that is, if $ae - 4c^2$ is $= -$, and the sign of c be opposite to that of a and e, or, what is the same thing, if ce be $= -$; but if these two conditions are not satisfied, then the values of ξ^2 will be imaginary, or else real and negative, and in either case the character will be $4ri$.

Article 282 (278)**.** The Bicorn curve has been constructed by means of the equation

$$(x^2 + 2xy - y^2)^2 + y^3(2x + y) = 0,$$

and the Plate shows the form of the curve with its six regions. The node-cusp at the origin is a singular point of a special character; and the other cusp is at infinity on the axis of x. The double tangent is the line $x + y = 0$, touching the curve at the two points determined by $x^2 + 2xy - y^2 = 0$, $x + y = \sqrt[4]{-\frac{1}{4}y^3(2x + y)}$.

Article No. 279. On the Geometrical Representation.

Article 283 (279)**.** The geometrical representation of the roots of a quintic equation is obtained by means of the canonical form. If in the plane we take a triangle ABC, and upon the sides thereof measure off distances proportional to x, y, z respectively, such that $x + y + z = 0$ (where account is taken of the signs), then if $x : y : z$ be proportional to the roots of the quintic equation, the locus of the point determined by these distances is a curve of the fifth class; and the different characters of the equation correspond to different configurations of this curve.

Article No. 280. Discussion of the Characters.

Article 284 (280)**.** The characters of a quintic equation are completely determined by the absolute invariants and the sign of I. The absolute invariants are the ratios of $J^3 : K^3 : L$ which are unaltered by any linear transformation; and the sign of I distinguishes between the characters $5r$ and $r+4i$ when the discriminant is positive and the quadrivariant negative. The different characters correspond to different regions of the discriminant-surface, separated by the nodal line and the cuspidal edge; and the complete discussion requires the consideration of the Bicorn curve and its six regions.

Article 285 (281)**.** The form of the Bicorn, so far as it is material for the discussion, is also shown in the Plate, fig. 3, and it thereby appears that it divides the plane into three regions on one side and three on the other. For the region P, we must have $J = -$, and for TU we must have $J = +$. Hence in connexion with the bicorn, considering the line $y = 0$, we have the six regions P, Q, R, S, T, U. It has just been seen that for P, Q, R, S the equations have $J = -$ and for T, U the equations have $J = +$; and the regions have determinate characters, which for R, S, and U (since here D is $= -$) is at once seen to be $3r + 2i$, and which, as will presently appear, is ascertained to be $5r$ for P and $r + 4i$ for Q and T.

Article 286 (282)**.** We have thus the Table

	D	J	$2^4L - J^2$		
P,	$D = +$,	$J = -$,	$2^4L - J^2 = +$	}	$5r$,
Q,	$D = +$,	$J = -$,	$2^4L - J^2 = -$	}	$r + 4i$,
T,	$D = +$,	$J = +$,	$2^4L - J^2 = +$	}	$r + 4i$,
R,	$D = -$,	$J = -$,	$2^4L - J^2 = \pm$	}	$3r + 2i$,
S,	$D = -$,	$J = -$,	$2^4L - J^2 = +$	}	$3r + 2i$,
U,	$D = -$,	$J = +$,	$2^4L - J^2 = +$	}	$3r + 2i$,

so that we have the kingdom $5r$ consisting of the single region P, the kingdom $r + 4i$ consisting of the regions Q and T, and the kingdom $3r + 2i$ consisting of the regions R, S, and U.

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Article 287 (283)**.** For a given equation if D is $= -$, the character is $3r + 2i$; if $D = +$, $J = +$, the character is $r + 4i$; if $D = +$, $J = -$, then, according as $2^4L - J^2$ is $= +$ or is $= -$ the character is $5r$ or $r + 4i$. But in the last case the distinction between the characters $5r$ and $r + 4i$ may be presented in a more general form, involving a parameter μ, arbitrary between certain limits. In fact drawing upwards from the origin, as in Plate, fig. 3, the lines $x - 2y = 0$ and $x + y = 0$, and between them any line whatever $x + \mu y = 0$, the point (x, y), assumed to lie in the region P or Q, will lie in the one or the other region according as it lies on the one side or the other side of the line in question, viz. in the region P if $x + \mu y$ is $= -$, in the region Q if $x + \mu y$ is $= +$. But we have

$$2^4L - J^2 + \mu JD$$

corresponds to $x + \mu y$, and J being by supposition negative, the sign of $2^4L - J^2 + \mu JD$ is opposite to that of $x + \mu y$. The region is thus P or Q according to the sign of $2^4L - J^2 + \mu JD$; and completing the enunciation, we have, finally, the following criteria for the number of real roots of a given quintic equation, viz.

If $D = -$, the character is $3r + 2i$,

If $D = +$, $J = +$, then it is $r + 4i$,

If $D = +$, $J = -$, then μ being any number at pleasure between the limits $+1$ and -2, both inclusive, if

$2^4L - J^2 + \mu JD = +$ the character is $5r$,

$2^4L - J^2 + \mu JD = -$, „ „ „ $r + 4i$.

Article 288 (284)**.** The characters $5r$ of the region P and $r+4i$ of the regions Q and T may be ascertained by means of the equation $(a,0,c,0,e,0 \mid x,1)^5=0$, that is

$$x(a\xi^4+10c\xi^2+e)=0;$$

there is always the real root $\xi=0$, and the equation will thus have the character $5r$ or $r+4i$ according as the reduced equation $a\xi^4+10c\xi^2+e=0$ has the character $4r$ or $4ri$. It is clear that (a,e) must have the same sign, for otherwise ξ^2 would have two real values, one positive, the other negative, and the character would be $2r+2i$. And (a,e) having the same sign, then the character will be $4r$, if ξ^2 has two real positive values, that is, if $ae-4c^2$ is $=-$, and the sign of c be opposite to that of a and e, or, what is the same thing, if ce be $=-$; but if these two conditions are not satisfied, then the values of ξ^2 will be imaginary, or else real and negative, and in either case the character will be $4ri$.

Article 289 (285)**.** Now, for the equation in question, putting in the Tables $b=d=f=0$, we find for the invariants

$$\begin{aligned}J&=16c^2e^2(ae-4c^2),\\D&=16c^2e^2\{(ae-4c^2)^2-8c^2e^2\},\\L&=a^4e^4,\end{aligned}$$

whence we can calculate the values of J, D, L for any given values of a, c, e. In particular, the equation $a\xi^4+10c\xi^2+e=0$ has the character $4r$ if $ae-4c^2=-$ and $c=-$, that is, if $J=+$ and $c=-$; but we have $D=+$ when $ae-4c^2=+$, or when $(ae-4c^2)^2-8c^2e^2=+$; and the sign of D being given, the character is determined as above.

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Article No. 286. On the Determination of the Character.

Article 290 (286)**.** The character of a quintic equation may be determined by the following process. First calculate the invariants J, D, L. Then if $D = -$, the character is $3r + 2i$. If $D = +$ and $J = +$, the character is $r + 4i$. If $D = +$ and $J = -$, then calculate $2^4 L - J^2$; if this is $+$, the character is $5r$; if $-$, the character is $r + 4i$. The case where $2^4 L - J^2 = 0$ is the critical case where the character changes from $5r$ to $r + 4i$, and corresponds to a point on the boundary between the regions P and Q.

Article 291 (287)**.** The discriminant D is an invariant of the degree 8, and may be expressed in terms of the fundamental invariants J, K, L. The expression is of the form

$$D = K^2 + \text{terms involving } J \text{ and } L,$$

and the complete expression is a complicated function of these invariants. The surface $D = 0$ is the discriminant-surface, and it divides the space of invariants into regions corresponding to the different characters.

Article 292 (288)**.** The nodal line of the discriminant-surface corresponds to the case where the quintic has two pairs of equal roots. This is the case where the catalecticant vanishes, and where the quintic can be expressed as the sum of three fifth powers. The triple point corresponds to the case of three equal roots, and lies on the nodal line. The different sections of the discriminant-surface by planes give rise to curves which have been studied by Sylvester and others.

Article 293 (289)**.** The complete discussion of the characters requires the consideration of the absolute invariants, that is, the ratios which are unaltered by any linear transformation. These absolute invariants may be taken to be $\dfrac{J^3}{K^3}$ and $\dfrac{L}{K^2}$, or any other independent ratios formed from J, K, L; and the discriminant being expressed as a function of these, the different regions of the surface correspond to the different characters.

Article No. 290. On the Special Cases.

Article 294 (290)**.** The special cases of the quintic equation arise when the discriminant vanishes, that is, when two or more roots are equal. The case of a single pair of equal roots corresponds to a point on the discriminant-surface; the case of two pairs of equal roots corresponds to a point on the nodal line; and the case of three equal roots corresponds to the triple point. The case of four equal roots or of five equal roots corresponds to higher singularities which are not in general considered in the theory of the quintic.

Article 295 (291)**.** When the discriminant vanishes, the character of the equation changes by the coalescence of two roots. If two real roots coalesce, the character changes from $5r$ to $3r + 2i$, or from $3r + 2i$ to $r + 4i$, according to the circumstances. If two imaginary roots coalesce, the character changes from $r + 4i$ to $3r + 2i$. The complete enumeration of all the possible cases requires the consideration of the different ways in which the roots may coalesce.

Article 296 (292)**.** The case of a double root is distinguished from the case of two pairs of double roots by the condition that the first minors of the catalecticant do not all vanish. For a double root, the catalecticant vanishes, but not all its first minors; for two pairs of double roots, the first minors all vanish, but not the second minors; and so on. The complete theory of these special cases is intimately connected with the theory of the canonical form and the theory of the discriminant.

Article 297 (293)**.** The discriminant of the quintic is a function of the degree 8 in the coefficients, and it may be expressed as the resultant of the quintic and its first derived function. The discriminant is also expressible in terms of the invariants J, K, L, and the expression is a complicated function involving these invariants in various combinations. The discriminant is the most important invariant for the determination of the character of the equation.

Article No. 294. On the Application to Numerical Equations.

Article 298 (294)**.** The application of the preceding theory to numerical equations is effected by calculating the invariants J, D, L for the given equation, and then determining the character by the criteria above established. The calculation of the invariants is a process of considerable labour, but it may be simplified by the use of tables or by special devices. For equations with numerical coefficients, the invariants may be calculated directly from the coefficients, and the character determined without solving the equation.

Article 299 (295)**.** The invariants J, K, L may be calculated by the formulae given in the memoir, or by any other equivalent formulae. The discriminant D is then calculated from these invariants, and the character determined by the signs of D, J, and $2^4L - J^2$. The process is completely definite, and gives a direct determination of the number of real and imaginary roots without the necessity of solving the equation.

Article 300 (296)**.** The theory of the characters of a quintic equation is analogous to the theory of the discriminant of a quadratic or cubic equation. For a quadratic equation, the discriminant determines whether the roots are real or imaginary; for a cubic equation, the discriminant determines whether the roots are all real or one real and two imaginary; and for a quintic equation, the discriminant together with the other invariants determines the number of real and imaginary roots in each of the possible cases.

Article 301 (297)**.** The complete determination of the character of a quintic equation requires the knowledge of the absolute invariants and the sign of the skew invariant I. The absolute invariants determine the ratios of the coefficients of the canonical form, and the sign of I determines the particular character within each class. The theory is thus complete, and gives a definite answer for every quintic equation.

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Article No. 298. Summary of Results.

Article 302 (298). The results obtained in the present memoir may be summarized as follows:

1. The covariants of the quintic up to the degree 6 have been completely calculated and tabulated.
2. The syzygies between these covariants have been determined.
3. The canonical form $ax^5 + by^5 + cz^5$ $(x + y + z = 0)$ has been discussed, and the invariants and covariants expressed in terms of the coefficients (a, b, c).
4. The 18-thic invariant I has been expressed as a function of the invariants J, K, L, and also in terms of the roots.
5. The characters of the quintic equation have been completely determined by means of the invariants and the Bicorn curve.
6. The discriminant-surface and its singularities have been discussed.

Article 303 (299). The memoir concludes with a discussion of the general theory of quantics, and of the methods applicable to the investigation of their invariants and covariants. The methods employed are those which have been developed in the preceding memoirs, and the results are presented in a form suitable for application to particular cases. The theory of the quintic is thus brought to a state of completeness which renders it available for all the purposes of algebraical and geometrical investigation.

Article 304 (300). The Tables of covariants which have been given in the memoir serve to exhibit the complete system of covariants up to the degree 6, and they may be used for the calculation of any particular covariant which may be required. The syzygies serve to reduce the number of independent covariants, and to express the relations between them. The theory is thus presented in a compact and manageable form.

Article 305 (301). The discussion of the characters of the quintic equation by means of the Bicorn curve is a novel feature of the memoir, and it serves to connect the algebraical theory with the geometrical theory of curves. The Bicorn curve is a curve of the fourth order with a node-cusp and a cusp at infinity, and it divides the plane into six regions corresponding to the different characters of the quintic equation. The criteria for the determination of the character are expressed in a simple form involving the invariants J, D, L.

Article 306 (302). The theory of the quintic equation here developed is applicable to the solution of problems in algebra, geometry, and the theory of equations. The determination of the character of a given equation is a problem of frequent occurrence, and the methods here given render it completely determinate. The theory of the canonical form is also of importance in connexion with the reduction of the general equation to the simplest form, and the expression of the invariants and covariants in terms of the coefficients of the canonical form is a result of considerable value.

Article 307 (303). The memoir contains also a discussion of the superimaginary and subimaginary transformations, and of their effect upon the character of the equation. It is shown that the imaginary transformation $x : y$ into $iX : Y$ changes the sign of the quadrivariant J, and that the real equations which correspond to given values of the absolute invariants must be derivable one from another by real transformations. The absolute invariants, together with the sign of J, thus constitute a complete system of auxiliars.

Article 308 (304). The application of the theory to the auxiliars of a quintic is given in Article 327. It is shown that for a quintic equation $(a, \dots \mid x, y)^5 = 0$, after any real transformation whatever, the only cases in which an imaginary transformation produces a real equation of an altered character are when the equation is of the form $(a', 0, c', 0, e', 0 \mid x', y')^5 = 0$ $(c' \neq 0)$, or when it is of the form $(a', 0, 0, 0, e', 0 \mid x', y')^5 = 0$. In the latter case the transformation x', y' into $X\sqrt{-1}, Y$ gives the real equation $X(a'X^4 - 5e'Y^4) = 0$.

Article 309 (305). For the form $(a', 0, 0, 0, e', 0 \mid x, y)^5$, and consequently for the form $(a, \dots \mid x, y)^5$ from which it is derived, we have $J = 0$; this case is therefore excluded from consideration. The remaining case is $(a', 0, c', 0, e', 0 \mid x', y')^5 = 0$, which is by the imaginary transformation $x' : y'$ into $iX : Y$ converted into $(a', 0, -c', 0, e', 0 \mid X, Y)^5 = 0$; for the first of the two forms we have $J = 16a'c'e'^2$, and for the second of the two forms $J = -16a'c'e'^2$, that is, the two values of J have opposite signs. Hence considering an equation $(a, b, c, d, e, f \mid x, y)^5 = 0$ for which J is not $= 0$, whenever this is by an imaginary transformation converted into a real equation, the sign of J is reversed; and it follows that, given the values of the absolute invariants and the value of J (or what is sufficient, the sign of J), the different real equations which correspond to these data must be derivable one from another by real transformations, and must consequently have a determinate character; that is, the Absolute Invariants, and J, constitute a system of auxiliars.

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Annex. Analytical Theorem in relation to a Binary Quantic of any Order.

Article 310 (306)**.** The foregoing theory of the superimaginary and subimaginary transformations leads to the following general theorem. If $(a, \dots \mid x, y)^n = 0$ be a binary quantic of any order n, and if by any linear transformation whatever the equation is converted into a real equation of the same or of a different character, then the transformation may be reduced to one of the following forms:

1. A real transformation, which preserves the character of the equation;
2. An imaginary transformation of the form $x : y$ into $kX : Y$, where k^p is real for some divisor p of n, which may or may not alter the character;
3. A superimaginary transformation, which changes the character in a determinate manner.

Article 311 (307)**.** The theorem may be established by considering the general linear transformation

$$x = \alpha X + \beta Y, \quad y = \gamma X + \delta Y,$$

where α, β, γ, δ are arbitrary quantities. If the transformed equation is to be real, then the coefficients of the transformed equation must all be real; and this condition restricts the possible values of α, β, γ, δ. The complete discussion of the cases leads to the classification above given.

Article 312 (308)**.** It is to be observed that the superimaginary transformation is only possible when the equation has a certain special form; in particular, it is necessary that the equation should be expressible in terms of x^p and y^p for some divisor p of n. When this condition is satisfied, the transformation $x : y$ into $iX : Y$ (where $i = \sqrt{-1}$) gives a real equation, and the character is altered in a determinate manner. The cases in which this occurs for the quintic equation have been discussed in the preceding articles.

Article 313 (309)**.** The subimaginary transformation is the transformation $x : y$ into $kX : Y$, where k^p is real for some divisor p of n. If p is odd, the transformation is equivalent to a real transformation, and the character is unaltered. If p is even and k^p is positive, the same is true. But if p is even and k^p is negative, then the transformation is not equivalent to any real transformation, and the character of the equation may be altered. The alteration of character in this case has been discussed for the quintic equation, and the general theory is analogous.

Article 314 (310)**.** The theorem may be further extended to the case of a system of binary quantics, and to the case of ternary and higher quantics. The general principle is that the imaginary transformations which produce real equations are of a limited class, and that their effect upon the character of the equation is determinate. The theory of these transformations is thus a branch of the general theory of invariants, and it has important applications in algebra and geometry.

Article 315 (311)**.** The application of the theorem to the theory of equations is immediate. Given a binary quantic of any order, the theorem enables us to determine whether the equation can be transformed into a real equation of a different character by an imaginary transformation, and if so, what is the nature of the transformation and the effect upon the character. The theorem also enables us to classify the different transformations which leave the character unaltered, and to distinguish them from those which alter it.

Article 316 (312)**.** The proof of the theorem depends upon the theory of the roots of unity, and upon the properties of the linear transformation of binary forms. The general linear transformation involves four arbitrary parameters, and the condition that the transformed equation shall be real imposes certain restrictions upon these parameters. The complete discussion of the cases leads to the classification of transformations into real, subimaginary, and superimaginary, and to the determination of the effect of each class of transformations upon the character of the equation.

Article 317 (313)**.** The memoir concludes with a discussion of the general theory of the transformations of binary quantics, and of their application to the theory of equations. The results obtained are of a general character, and they serve to unify and complete the theory of the invariants and covariants of binary forms. The theory of the quintic equation is a particular case of the general theory, and the results obtained for the quintic are illustrative of the general principles.

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Article No. 314. On the Theory of Transformations.

Article 318 (314)**.** The theory of the transformation of binary quantics is a fundamental part of the theory of invariants. The general linear transformation

$$x = \alpha X + \beta Y, \quad y = \gamma X + \delta Y,$$

where $\alpha\delta - \beta\gamma \neq 0$, transforms any binary quantic into another binary quantic of the same order; and the invariants and covariants of the original form are transformed into invariants and covariants of the new form. The theory of these transformations has been developed by Cayley, Sylvester, and others, and it has important applications in algebra and geometry.

Article 319 (315)**.** The transformation is said to be real when the coefficients α, β, γ, δ are all real; and in this case, if the original equation is real, the transformed equation is also real. The transformation is said to be imaginary when some or all of the coefficients are imaginary; and in this case, even if the original equation is real, the transformed equation may be imaginary. The cases in which an imaginary transformation produces a real equation are of special interest, and they have been discussed in the preceding articles.

Article 320 (316)**.** The transformation $x : y$ into $iX : Y$ is a particular case of the superimaginary transformation. If the original equation is $(a, \ldots \mid x, y)^n = 0$, the transformed equation is $(a, \ldots \mid iX, Y)^n = 0$; and if the original equation contains only even powers of x and y, or only powers which are multiples of some divisor of n, then the transformed equation will be real. The effect of the transformation upon the character of the equation is to change the sign of certain invariants, and the complete discussion of this effect has been given for the case of the quintic.

Article 321 (317)**.** The transformation $x : y$ into $kX : Y$, where k^p is real for some divisor p of n, is the subimaginary transformation. If p is odd, the transformation is equivalent to a real transformation; if p is even and k^p is positive, the same is true; but if p is even and k^p is negative, the transformation is not equivalent to any real transformation, and the character of the equation may be altered. The cases in which this occurs have been discussed for the quintic, and the general theory is the same for quantics of any order.

Article 322 (318)**.** The theory of the transformation of binary quantics has important applications in the theory of equations. The determination of the character of an equation, that is, the number of its real and imaginary roots, is a problem of fundamental importance; and the theory of transformations enables us to determine the character without solving the equation. The invariants and covariants of the quantic furnish the means of this determination, and the theory of transformations serves to connect the different forms of the equation and to exhibit the relations between their invariants.

Article 323 (319)**.** The memoir concludes with a general discussion of the theory of binary quantics and their transformations. The results obtained are of a comprehensive character, and they serve to complete the theory which has been developed in the preceding memoirs. The theory of the quintic equation is a particular application of the general theory, and the methods employed are applicable to quantics of any order. The memoir thus forms a contribution to the general theory of algebraical forms, and it illustrates the power and generality of the methods of invariant theory.

Article 324 (320)**.** The following is a brief recapitulation of the principal results of the memoir:

1. The complete system of covariants of the quintic up to the degree 6 has been calculated and tabulated.
2. The syzygies between these covariants have been determined.
3. The canonical form $ax^5 + by^5 + cz^5$ $(x + y + z = 0)$ has been discussed.
4. The 18-thic invariant has been expressed in terms of the invariants J, K, L.
5. The characters of the quintic equation have been determined by means of the Bicorn curve.
6. The theory of superimaginary and subimaginary transformations has been developed.
7. The application to the auxiliars of a quintic has been given.

Article 325 (321)**.** The theory of the characters of the quintic equation is thus brought to a state of completeness. The discriminant-surface, with its nodal line and triple point, furnishes a complete classification of the possible characters; and the criteria for the determination of the character are expressed in terms of the invariants J, D, L, which can be calculated for any given equation. The theory is analogous to the theory of the discriminant of the quadratic and cubic equations, and it extends those theories to the case of the quintic.

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Article 326 (322)**.** The geometrical representation of the roots of a quintic equation by means of the canonical form $ax^5 + by^5 + cz^5 = 0$ $(x + y + z = 0)$ is a representation of considerable interest. If in the plane we take a triangle ABC, and upon the sides thereof measure off distances x, y, z such that $x + y + z = 0$, then the locus of the point so determined, when $x : y : z$ are the roots of the quintic equation, is a curve of the fifth class. The different characters of the equation correspond to different configurations of this curve, and the discriminant-surface corresponds to the cases where the curve acquires singularities.

Article 327 (323)**.** The theory of the transformations of binary quantics is thus seen to be a theory of wide extent and great generality. The methods employed are applicable not only to the quintic but to quantics of any order; and the results obtained for the quintic are typical of the results which may be expected for higher quantics. The theory of invariants and covariants, which is the foundation of the whole, is a theory of the most comprehensive character, and it has applications in every branch of algebra and geometry.

Article 328 (324)**.** There remains only the subimaginary transformation, viz. this has been reduced to $x : y$ into $kX : Y$, the transformed equation is $(a, \dots \mid kX, Y)^n = 0$, and this will be a real equation if some power k^p of k (p not greater than n) be real, and if the equation $(a, \dots \mid x, y)^n = 0$ contain only terms wherein the index of x (or that of y) is a multiple of p. Assuming that it is the index of y which is a multiple, the form of the equation is in fact $(x^p, y^p)^m = 0$, $(n = mp + \alpha)$, and the transformed equation is $X^\alpha(k^pX^p, Y^p)^m = 0$, which is a real equation.

Article 329 (325)**.** It is to be observed that if p be odd, then writing $k^p = K$ (K real) and taking k' the real p-th root of K, then the very same transformed equation would be obtained by the real transformation $x : y$ into $k'X : Y$; so that the equation obtained by the imaginary transformation, being also obtainable by a real transformation, has the same character as the original equation.

Article 330 (326)**.** Similarly if p be even, if K be real and positive, the equation $k^p = K$ has a real root k' which may be substituted for the imaginary k, and the transformed equation will have the same character as the original equation; but if K be negative, say $K = -\Lambda$ (as may be assumed without loss of generality), then there is no real transformation equivalent to the imaginary transformation, and the equation given by the imaginary transformation has not of necessity the same character as the original equation; and there are in fact cases in which the character is altered. Thus if $p = 2$, and the original equation be $x(x^2, y^2)^m = 0$, or $(x^2, y^2)^m = 0$, then making the transformation $x : y$ into $iX : Y$, the transformed equation will be $X(X^2, -Y^2)^m = 0$ or $(X^2, -Y^2)^m = 0$, giving imaginary roots $X^2 + aY^2 = 0$ corresponding to real roots $x^2 - ay^2 = 0$.

Article No. 327. Application to the auxiliars of a quintic.

Article 331 (327)**.** Applying what precedes to a quintic equation $(a, \ldots \mid x, y)^5 = 0$, this after any real transformation whatever will assume the form $(a, \ldots \mid x', y')^5 = 0$; and the only cases in which we can have an imaginary transformation producing a real equation of an altered character is when this equation is $(a', 0, c', 0, e', 0 \mid x', y')^5 = 0$ $(c' \neq 0)$, or when it is $(a', 0, 0, 0, e', 0 \mid x', y')^5 = 0$, viz. when it is $x'(a'x'^4 + 10c'x'^2y'^2 + 5e'y'^4) = 0$, or $x'^4(a'x'^2 + 5e'y'^2) = 0$. In the latter case the transformation x', y' into $X\sqrt{-1}, Y$ gives the real equation $X(a'X^4 - 5e'Y^4) = 0$. I observe however that for the form $(a', 0, 0, 0, e', 0 \mid x, y)^5$, and consequently for the form $(a, \ldots \mid x, y)^5$ from which it is derived we have $J = 0$; this case is therefore excluded from consideration. The remaining case is $(a', 0, c', 0, e', 0 \mid x', y')^5 = 0$, which is by the imaginary transformation $x' : y'$ into $iX : Y$ converted into $(a', 0, -c', 0, e', 0 \mid X, Y)^5 = 0$; for the first of the two forms we have $J = 16a'c'e'^2$, and for the second of the two forms $J = -16a'c'e'^2$, that is, the two values of J have opposite signs. Hence considering an equation $(a, b, c, d, e, f \mid x, y)^5 = 0$ for which J is not $= 0$, whenever this is by an imaginary transformation converted into a real equation, the sign of J is reversed; and it follows that, given the values of the absolute invariants and the value of J (or what is sufficient, the sign of J), the different real equations which correspond to these data must be derivable one from another by real transformations, and must consequently have a determinate character; that is, the Absolute Invariants, and J, constitute a system of auxiliars.

Article No. 328. On the geometrical theory of the quintic.

Article 332 (328)**.** The geometrical theory of the quintic equation is closely connected with the algebraical theory. The equation $(a, b, c, d, e, f \mid x, y)^5 = 0$ may be regarded as determining a range of five points on a straight line; and the invariants and covariants of the quantic have geometrical interpretations in connexion with this range of points. The discriminant vanishes when two of the points coincide; and the different characters of the equation correspond to different dispositions of the five points upon the real line.

Article 333 (329)**.** The canonical form $ax^5 + by^5 + cz^5 = 0$ $(x + y + z = 0)$ has a geometrical interpretation in terms of a triangle. If ABC be a triangle, and if upon the sides BC, CA, AB we measure off distances x, y, z such that $x + y + z = 0$, then any point in the plane determines a system of values of x, y, z; and conversely, any system of values of x, y, z satisfying this condition determines a point in the plane. The equation $ax^5 + by^5 + cz^5 = 0$ thus represents a curve of the fifth order in the plane of the triangle; and the roots of the quintic equation correspond to the intersections of this curve with the sides of the triangle.

Article 334 (330)**.** The theory of the characters of the quintic equation is thus seen to have a geometrical aspect. The different characters correspond to different configurations of the five points; and the discriminant-surface, with its nodal line and triple point, corresponds to the cases where the configuration acquires singularities. The theory of the Bicorn curve is a particular aspect of this geometrical theory, and it serves to connect the algebraical and geometrical points of view.

Article 335 (331)**.** The memoir concludes with a discussion of the general principles which underlie the theory of binary quantics and their invariants. The theory is presented as a systematic development of ideas which have their origin in the work of earlier mathematicians, but which have been greatly extended and generalized by the labours of Cayley, Sylvester, and their successors. The theory of the quintic equation is a particular but important application of these general principles, and it illustrates the power and beauty of the method of invariants.

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Article No. 332. On the higher covariants of the quintic.

Article 336 (332)**.** The covariants of the quintic which have been tabulated in the present memoir extend up to the degree 6. The complete system of covariants of the quintic is known to contain twenty-three members, including the quintic itself; and these covariants are of various degrees and orders. The calculation of the higher covariants is a work of considerable labour, but the methods employed in the present memoir are applicable to their determination. The syzygies between the covariants also become more numerous and more complicated as the degree increases.

Article 337 (333)**.** The sextic covariants M and N, which are given in Tables Nos. 83 and 84, are of special interest on account of their connexion with the discriminant and the other invariants. The covariant M is the Hessian of three times the cubic covariant D, and the covariant N is one-sixth of the Jacobian of the quadric covariant B and the quartic covariant H. These covariants, together with those previously calculated, form a complete system up to the degree 6.

Article 338 (334)**.** The theory of the higher covariants is connected with the theory of the higher singularities of the discriminant-surface. Each covariant has a geometrical significance in connexion with the curve represented by the quantic; and the vanishing of a covariant corresponds to a special configuration of the points of the range. The complete theory of these singularities is a subject of great complexity, but the principles upon which it depends have been established in the present and preceding memoirs.

Article 339 (335)**.** The memoir thus contributes to the theory of the quintic equation in several directions. It completes the theory of the covariants up to the degree 6; it establishes the criteria for the determination of the character of the equation; it develops the theory of the canonical form; and it discusses the effect of imaginary transformations upon the character. These results, taken together with those of the preceding memoirs, form a complete and systematic theory of the binary quintic.

Article 340 (336)**.** The superimaginary transformation is the transformation

$$x + iy : x - iy \text{ into } (\cos\phi + i\sin\phi)(X + iY) : (\cos\phi - i\sin\phi)(X - iY),$$

which may also be written

$$x + iy : x - iy \text{ into } e^{i\phi}(X + iY) : e^{-i\phi}(X - iY).$$

The transformed equation is

$$(a + \beta i, \ldots \mid (\cos\phi + i\sin\phi)(X + iY),\ (\cos\phi - i\sin\phi)(X - iY))^n = 0.$$

The left-hand side consists of terms such as $(X^2 + Y^2)^{n-s}$ into

$$(\gamma + \delta i)(\cos s\phi + i\sin s\phi)(X + iY)^s + (\gamma - \delta i)(\cos s\phi - i\sin s\phi)(X - iY)^s,$$

viz. the expression last written down is

$$\begin{aligned}= (\gamma\cos s\phi - \delta\sin s\phi)\,&[(X + iY)^s + (X - iY)^s] \\ &- (\gamma\sin s\phi + \delta\cos s\phi)\,[(X + iY)^s - (X - iY)^s]/(i),\end{aligned}$$

and hence for the expressions to be real, we must have $\tan s\phi =$ real; and observing that if $\tan s\phi$ be equal to any given real quantity whatever, then the values of $\tan\phi$ are all of them real, and that $\tan\phi$ real gives $\cos\phi$ and $\sin\phi$ each of them real, and therefore also ϕ real, it appears that the transformed equation is only real for the transformation

$$x + iy : x - iy \text{ into } (\cos\phi + i\sin\phi)(X + iY) : (\cos\phi - i\sin\phi)(X - iY),$$

where ϕ is real; and hence neither in the natural transformation nor in that of the superimaginary transformation can we have an imaginary transformation leading to a real equation.

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Article 341 (337)**.** The theory of the superimaginary transformation is thus seen to lead to the conclusion that no new case arises from it. The transformed equation is only real when ϕ is real, and in this case the transformation is equivalent to a real transformation. The superimaginary transformation thus does not give any case in which an imaginary transformation produces a real equation of an altered character.

Article 342 (338)**.** The subimaginary transformation remains to be considered. This is the transformation $x : y$ into $kX : Y$, where k^p is real for some divisor p of n. If p is odd, the transformation is equivalent to a real transformation, and the character is unaltered. If p is even and k^p is positive, the same is true. But if p is even and k^p is negative, then the transformation is not equivalent to any real transformation, and the character of the equation may be altered.

Article 343 (339)**.** The case $p = 2$, $n = 4$ or $n = 5$ is the case which has been discussed in detail for the quintic. The original equation $(x^2, y^2)^m = 0$ is transformed into $(X^2, -Y^2)^m = 0$, and the character is altered. For $m = 1$, the equation $x^2 - ay^2 = 0$ (two real roots) is transformed into $X^2 + aY^2 = 0$ (two imaginary roots); and for higher values of m the alteration of character is similar.

Article 344 (340)**.** The complete discussion of the subimaginary transformation for the quintic has been given in Articles 324–327, and the result is that the only cases in which an imaginary transformation alters the character are the cases there enumerated. The general theory of these transformations for quantics of any order follows the same lines, and leads to similar results.

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Article 345 (341)**.** The theory of the transformations of binary quantics is thus complete. The general linear transformation involves four arbitrary parameters, and the condition that the transformed equation shall be real imposes certain restrictions upon these parameters. The complete discussion leads to the classification of transformations into real, subimaginary, and superimaginary; and the effect of each class upon the character of the equation has been determined.

Article 346 (342)**.** The application of this theory to the problem of the determination of the character of a binary quantic is immediate. Given the coefficients of the quantic, we calculate the invariants; and from the values of the invariants we determine the character by the criteria which have been established. The process is completely definite, and it does not require the solution of the equation. The theory is thus of great practical value, as well as of theoretical interest.

Article 347 (343)**.** The memoir concludes with a general survey of the theory of binary quantics. The theory is presented as a systematic development, founded upon the fundamental notions of invariants and covariants, and extending to the most general results. The theory of the quintic is a particular but important application, and it has been developed in detail in the present and preceding memoirs. The results obtained are of permanent value, and they form a contribution to the science of algebra which will not easily be superseded.

Article 348 (344)**.** The following is a brief recapitulation of the principal results:

1. The complete system of covariants of the quintic up to the degree 6.
2. The determination of the syzygies between these covariants.
3. The expression of the 18-thic invariant in terms of the invariants J, K, L.
4. The criteria for the determination of the character of a quintic equation.
5. The theory of the canonical form $ax^5 + by^5 + cz^5$ $(x + y + z = 0)$.
6. The theory of superimaginary and subimaginary transformations.
7. The proof that the Absolute Invariants and J constitute a system of auxiliars.

Article 349 (345). The discriminant of the quintic is a function of the degree 8 in the coefficients, and it may be expressed in terms of the invariants J, K, L. The expression is

$$D = K^2 + \text{terms involving } J \text{ and } L,$$

and the complete expression is a complicated function of these invariants. The discriminant-surface $D = 0$ divides the space of invariants into regions corresponding to the different characters of the equation.

Article 350 (346). The nodal line of the discriminant-surface corresponds to the case of two pairs of equal roots. On this line, the discriminant and its first derived functions all vanish; and the nodal line itself has a triple point corresponding to the case of three equal roots. The different sections of the discriminant-surface by planes give rise to curves which have been studied by Sylvester and others.

Article 351 (347). The characters of the quintic equation are completely determined by the absolute invariants together with the sign of the skew invariant I. The absolute invariants are functions of J, K, L which are unaltered by any linear transformation; and the sign of I distinguishes between characters which have the same absolute invariants. The criteria for the determination of the character have been given in a form suitable for application to numerical equations.

Article 352 (348). The theory of the transformations of binary quantics has been applied to show that the absolute invariants, together with the sign of J, constitute a complete system of auxiliars. The proof depends upon the classification of transformations into real, subimaginary, and superimaginary, and upon the determination of the effect of each class of transformations upon the character of the equation. The result is that, given the absolute invariants and the sign of J, the character of the equation is completely determined.

Article 353 (349)**.** There remains only the subimaginary transformation, viz. this has been reduced to $x : y$ into $kX : Y$, the transformed equation is $(a, \dots \mid kX, Y)^n = 0$, and this will be a real equation if some power k^p of k (p not greater than n) be real, and if the equation $(a, \dots \mid x, y)^n = 0$ contain only terms wherein the index of x (or that of y) is a multiple of p. Assuming that it is the index of y which is a multiple, the form of the equation is in fact $(x^p, y^p)^m = 0$, $(n = mp + \alpha)$, and the transformed equation is $X^\alpha(k^p X^p, Y^p)^m = 0$, which is a real equation.

Article 354 (350)**.** It is to be observed that if p be odd, then writing $k^p = K$ (K real) and taking k' the real p-th root of K, then the very same transformed equation would be obtained by the real transformation $x : y$ into $k'X : Y$; so that the equation obtained by the imaginary transformation, being also obtainable by a real transformation, has the same character as the original equation.

Article 355 (351)**.** Similarly if p be even, if K be real and positive, the equation $k^p = K$ has a real root k' which may be substituted for the imaginary k, and the transformed equation will have the same character as the original equation; but if K be negative, say $K = -\Lambda$ (as may be assumed without loss of generality), then there is no real transformation equivalent to the imaginary transformation, and the equation given by the imaginary transformation has not of necessity the same character as the original equation; and there are in fact cases in which the character is altered. Thus if $p = 2$, and the original equation be $x(x^2, y^2)^m = 0$, or $(x^2, y^2)^m = 0$, then making the transformation $x : y$ into $iX : Y$, the transformed equation will be $X(X^2, -Y^2)^m = 0$ or $(X^2, -Y^2)^m = 0$, giving imaginary roots $X^2 + aY^2 = 0$ corresponding to real roots $x^2 - ay^2 = 0$.

Article 356 (352)**.** Applying what precedes to a quintic equation $(a, \dots \mid x, y)^5 = 0$, this after any real transformation whatever will assume the form $(a, \dots \mid x', y')^5 = 0$; and the only cases in which we can have an imaginary transformation producing a real equation of an altered character is when this equation is $(a', 0, c', 0, e', 0 \mid x', y')^5 = 0$ $(c' \neq 0)$, or when it is $(a', 0, 0, 0, e', 0 \mid x', y')^5 = 0$, viz. when it is $x'(a'x'^4 + 10c'x'^2y'^2 + 5e'y'^4) = 0$, or $x'^4(a'x'^2 + 5e'y'^2) = 0$. In the latter case the transformation x', y' into $X\sqrt{-1}, Y$ gives the real equation $X(a'X^4 - 5e'Y^4) = 0$.

Article 357 (353)**.** The different real equations which correspond to these data must be derivable one from another by real transformations, and must consequently have a determinate character; that is, the Absolute Invariants, and J, constitute a system of auxiliars.

Annex. Analytical Theorem in relation to a Binary Quantic of any Order.

Article 358 (354)**.** The foregoing theory of the superimaginary and subimaginary transformations leads to the following general theorem. If $(a, \dots \mid x, y)^n = 0$ be a binary quantic of any order n, and if by any linear transformation whatever the equation is converted into a real equation of the same or of a different character, then the transformation may be reduced to one of the following forms:

1. A real transformation, which preserves the character of the equation;
2. An imaginary transformation of the form $x : y$ into $kX : Y$, where k^p is real for some divisor p of n, which may or may not alter the character;
3. A superimaginary transformation, which changes the character in a determinate manner.

Article 359 (355)**.** The theorem may be established by considering the general linear transformation

$$x = \alpha X + \beta Y, \quad y = \gamma X + \delta Y,$$

where α, β, γ, δ are arbitrary quantities. If the transformed equation is to be real, then the coefficients of the transformed equation must all be real; and this condition restricts the possible values of α, β, γ, δ. The complete discussion of the cases leads to the classification above given.

Article 360 (356)**.** It is to be observed that the superimaginary transformation is only possible when the equation has a certain special form; in particular, it is necessary that the equation should be expressible in terms of x^p and y^p for some divisor p of n. When this condition is satisfied, the transformation $x : y$ into $iX : Y$ (where $i = \sqrt{-1}$) gives a real equation, and the character is altered in a determinate manner. The cases in which this occurs for the quintic equation have been discussed in the preceding articles.

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Article 361 (357)**.** The subimaginary transformation is the transformation $x : y$ into $kX : Y$, where k^p is real for some divisor p of n. If p is odd, the transformation is equivalent to a real transformation, and the character is unaltered. If p is even and k^p is positive, the same is true. But if p is even and k^p is negative, then the transformation is not equivalent to any real transformation, and the character of the equation may be altered. The alteration of character in this case has been discussed for the quintic equation, and the general theory is analogous.

Article 362 (358)**.** The theorem may be further extended to the case of a system of binary quantics, and to the case of ternary and higher quantics. The general principle is that the imaginary transformations which produce real equations are of a limited class, and that their effect upon the character of the equation is determinate. The theory of these transformations is thus a branch of the general theory of invariants, and it has important applications in algebra and geometry.

Article 363 (359)**.** The application of the theorem to the theory of equations is immediate. Given a binary quantic of any order, the theorem enables us to determine whether the equation can be transformed into a real equation of a different character by an imaginary transformation, and if so, what is the nature of the transformation and the effect upon the character. The theorem also enables us to classify the different transformations which leave the character unaltered, and to distinguish them from those which alter it.

Article 364 (360)**.** The proof of the theorem depends upon the theory of the roots of unity, and upon the properties of the linear transformation of binary forms. The general linear transformation involves four arbitrary parameters, and the condition that the transformed equation shall be real imposes certain restrictions upon these parameters. The complete discussion of the cases leads to the classification of transformations into real, subimaginary, and superimaginary, and to the determination of the effect of each class of transformations upon the character of the equation.

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Article 365 (361). The memoir concludes with a discussion of the general theory of the transformations of binary quantics, and of their application to the theory of equations. The results obtained are of a general character, and they serve to unify and complete the theory of the invariants and covariants of binary forms. The theory of the quintic equation is a particular case of the general theory, and the results obtained for the quintic are illustrative of the general principles.

Article 366 (362). The following is a brief recapitulation of the principal results of the memoir:

1. The covariants of the quintic up to the degree 6 have been completely calculated and tabulated.
2. The syzygies between these covariants have been determined.
3. The canonical form $ax^5 + by^5 + cz^5$ $(x + y + z = 0)$ has been discussed, and the invariants and covariants expressed in terms of the coefficients (a, b, c).
4. The 18-thic invariant I has been expressed as a function of the invariants J, K, L, and also in terms of the roots.
5. The characters of the quintic equation have been completely determined by means of the invariants and the Bicorn curve.
6. The discriminant-surface and its singularities have been discussed.
7. The theory of superimaginary and subimaginary transformations has been developed.
8. The application to the auxiliars of a quintic has been given.

Article 367 (363). The theory of the characters of the quintic equation is thus brought to a state of completeness. The discriminant-surface, with its nodal line and triple point, furnishes a complete classification of the possible characters; and the criteria for the determination of the character are expressed in terms of the invariants J, D, L, which can be calculated for any given equation. The theory is analogous to the theory of the discriminant of the quadratic and cubic equations, and it extends those theories to the case of the quintic.

Article 368 (364). The geometrical representation of the roots of a quintic equation by means of the canonical form $ax^5 + by^5 + cz^5 = 0$ $(x + y + z = 0)$ is a representation of considerable interest. If in the plane we take a triangle ABC, and upon the sides thereof measure off distances x, y, z such that $x + y + z = 0$, then the locus of the point so determined, when $x : y : z$ are the roots of the quintic equation, is a curve of the fifth class. The different characters of the equation correspond to different configurations of this curve, and the discriminant-surface corresponds to the cases where the curve acquires singularities.

Article 369 (365). The theory of the transformations of binary quantics is thus seen to be a theory of wide extent and great generality. The methods employed are applicable not only to the quintic but to quantics of any order; and the results obtained for the quintic are typical of the results which may be expected for higher quantics. The theory of invariants and covariants, which is the foundation of the whole, is a theory of the most comprehensive character, and it has applications in every branch of algebra and geometry.

Article 370 (366). The memoir thus forms a contribution to the general theory of algebraical forms which is of permanent value. The results obtained are not only of theoretical interest, but they are also of practical utility in the solution of problems relating to the theory of equations. The determination of the character of a quintic equation, without solving it, is a problem which is completely solved by the methods of the present memoir; and the theory of the canonical form and the invariants furnishes the means of this determination.

Article 371 (367). The theory of the higher covariants and the higher singularities of the discriminant-surface remains for future investigation. The principles which have been established in the present and preceding memoirs are sufficient for the determination of the covariants up to any given degree, and for the discussion of the corresponding singularities; but the actual calculation of these covariants and the discussion of their properties is a work of great labour, which can only be undertaken in connexion with particular problems. The theory is thus left in a state in which it is ready for further development and application.

Article 372 (368). In conclusion, the author expresses his indebtedness to Professor Sylvester, whose researches on the quintic equation have been the starting-point of much of the present memoir. The theory of the amphigenous surface, which Professor Sylvester has developed, is replaced in the present memoir by a more direct method; but the fundamental ideas are due to him, and the results obtained are in harmony with those which he has established by his own methods.

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Article 373 (369). The theory of the quintic equation is thus brought to a state of completeness which renders it available for all the purposes of algebraical and geometrical investigation. The memoir concludes with the expression of the hope that the methods employed and the results obtained may serve to advance the science of algebra, and to facilitate the solution of problems which have hitherto been regarded as of insuperable difficulty.

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406. On the Curves which satisfy Given Conditions

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Article 374 (1)**.** The present memoir is an attempt to lay the foundations of a theory of the curves which satisfy given conditions. The problem is, given a curve of given order, the equation of which contains a certain number of arbitrary parameters, to determine the number of such curves which satisfy given conditions. The conditions may be of various kinds; they may be conditions that the curve shall pass through given points, or touch given curves, or have certain singularities; and the problem is to determine the number of curves which satisfy any given system of such conditions.

Article 375 (2)**.** The theory of the conics which satisfy five conditions (for instance, passing through five points, or touching five lines) is well known; the number of such conics is in general $= 1$. For cubics, the theory is more complicated, but it has been treated by Chasles, Zeuthen, and others. The general theory for curves of any order has not hitherto been established, and the present memoir is an attempt to supply this want.

Article 376 (3)**.** The method employed is to regard the arbitrary parameters in the equation of the curve as the coordinates of a point in a space of the proper dimension; the conditions then correspond to loci in this space, and the problem is to determine the number of points of intersection of these loci. This method has been employed by Chasles for the case of conics, and it is capable of extension to curves of any order.

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Article 377 (4). A condition imposed upon a curve may be of various orders. A condition that the curve shall pass through a given point is a onefold condition; a condition that it shall touch a given curve is also a onefold condition; a condition that it shall have a node is a twofold condition; and so on. The number of conditions which determine a curve of given order is equal to the number of arbitrary parameters in its equation; and when the curve is subjected to a number of conditions equal to this number, the curve is in general completely determined.

Article 378 (5). If the parameters are taken to be the coordinates of a point in a space of ω dimensions, then a k-fold condition establishes a k-fold relation between the parameters, and the parametric point is situate on a k-fold locus. If we have conditions of manifoldness k, l, &c., and if $k + l +$ &c. $= \omega$, then the loci intersect in a point-system, and the number of points of this point-system is equal to the product of the orders of the several loci. The order of the k-fold locus is the number of points in which it is intersected by a general $(\omega - k)$-fold linear locus.

Article 379 (6). The conditions may, however, not be completely independent; that is, the loci may have a common locus of less than ω dimensions, and may besides intersect in a residual point-system. The number of proper solutions is then equal to the order of the point-system, less the number of points which give special solutions. The determination of the order of the residual point-system, and the deduction of the special solutions, is the chief difficulty of the problem.

Article 380 (7). Imagine a curve of given order, the equation of which contains m arbitrary parameters: to fix the ideas, it may be assumed that these enter into the equation rationally, so that the values of the parameters being given, the curve is uniquely determined. Suppose, as above, that the parameters are taken to be the coordinates of a point in m-dimensional space; so long as the curve is not subjected to any condition, the point in question, say the parametric point, is an arbitrary point in the m-dimensional space; but if the curve be subjected to a onefold, twofold, ... or k-fold condition, then we have a onefold, twofold, ... or k-fold relation between the parameters, and the parametric point is situate on a onefold, twofold, ... or k-fold locus accordingly: to each position of the parametric point on the locus there corresponds a curve satisfying the condition, that is, a solution of the condition. In the case where the condition is ω-fold, the locus is a point-system, and corresponding to each point of the point-system we have a solution of the condition; the number of solutions is equal to the number of points of the point-system.

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Article 381 (8). Considering the general case where the condition, and therefore also the locus, is k-fold, it is to be observed that every solution whatever, and therefore each special solution (if any), corresponds to some point on the k-fold locus we may therefore have on the k-fold locus what may be termed "special loci," viz. a special locus is a locus such that to each point thereof corresponds a special solution. A special locus may of course be a point-system, viz. there are in this case a determinate number of special solutions corresponding to the several points of this point-system. We may consider the other extreme case of a special k-fold locus, viz. the k-fold locus of the parametric point may break up into two distinct loci, the special k-fold locus, and another k-fold locus the several points whereof give the ordinary solutions we can in this case get ; rid of the special solutions by attending exclusively to the last-mentioned k-fold locus and regarding it as the proper locus of the parametric point. But if the special locus be a more than k-fold locus, that is, if it be not a part of the k-fold locus itself, but (as supposed in the first instance) a locus on this locus, then the special solutions cannot be thus got rid of: we have the k-fold locus of the parametric point, a locus such that to every point thereof there corresponds a proper solution, save and except that to the points lying on the special locus there correspond special or improper solutions. It is to be noticed that the special locus may be, but that is not in every case, a singular locus on the k-fold locus.

Article 382 (9). Suppose that the conditions to be satisfied by the curve are a k-fold condition, an l-fold condition, &c. of a total manifoldness $= \omega$. If the conditions are completely independent (that is, if the corresponding relations, ante, No. 5, are completely independent), we have a k-fold locus, an l-fold locus, &c., having no common locus other than the point-system of intersection, and the number of curves which satisfy the given conditions, or (as this has been before expressed) the number of solutions, is equal to the number of points of the point-system, or to the order of the point-system, viz. it is equal to the product of the orders of the loci which correspond to the several conditions respectively; among these we may however have special solutions, corresponding to points situate on the special loci upon any of the given loci; but when this is the case the number of these special solutions can be separately calculated, and the number of proper solutions is equal to the number obtained as above, less the number of the special solutions.

Article 383 (10). If, however, the given conditions are not completely independent (that is, if the corresponding relations are not completely independent), then the k-fold locus, the l-fold locus, &c. intersect in a common $(\omega - j)$-fold locus, and besides in a residual point-system. The several points of the $(\omega - j)$-fold locus give special solutions; in fact the very notion of the conditions being properly satisfied by a curve implies that the curve shall satisfy a true $(k + l + \&c.)$-fold, that is, a true ω-fold condition; the proper solutions are therefore comprised among the solutions given by the residual point-system, and the number of them is as before equal to the order of the point-system, or number of the points thereof, less the number of points which give special solutions: the order of the point-system is, as has been seen, equal to the product of the orders of the k-fold locus, the l-fold locus, &c., less a reduction depending on the nature of the common $(\omega - j)$-fold locus, and the difficulty is in general in the determination of the value of this reduction.

Article 384 (11). In all that precedes, the number of the parameters has been taken to be ω; but if the parameters are taken to be contained in the equation of the curve homogeneously, then the parameters before made use of are in fact the ratios of these homogeneous parameters; and using the term henceforward as referring to the homogeneous parameters, the numbers of the parameters will be $= \omega + 1$.

Article 385 (12). I assume also that the equation of the curve contains the parameters linearly; this being so, the condition that the curve shall pass through a given arbitrary point implies a linear relation between the parameters; and the condition that the curve shall pass through j given points, a j-fold linear relation between the parameters. It follows that the number of the curves which satisfy a given k-fold condition, and besides pass through $\omega - k$ given points, is equal to the order of the k-fold relation, or of the corresponding k-fold locus; and thus if we define the order of the k-fold condition to be the number of the curves in question, the condition, relation, and locus will be all of the same order, and in all that precedes we may (in place of the order of the relation or of the locus) speak of the order of the condition. Thus, subject to the modifications occasioned by common loci and special solutions as above explained, the order of the $(k + l + \&c.)$-fold condition made up of a k-fold condition, an l-fold condition, &c., is equal to the product of the orders of the component conditions; and in particular if $k + l + \&c. = \omega$, then the order of the ω-fold condition, or number of the solutions thereof, is equal to the product of the orders of the component conditions.

Article 386 (13). The conditions to be satisfied by the curve may be conditions of contact with a given curve or curves. In particular if the curve touch a given curve, the parametric point is then situate on a onefold locus. It is to be noticed in reference hereto that if the given curve have nodes or cusps, then we have special solutions, viz. if the sought for curve passes through a node or a cusp of the given curve and each such node or cusp gives rise to a special onefold locus, presenting itself in the first instance as a factor of the onefold locus of the parametric point; this is, however, a case where the special locus is of the same manifoldness as the general locus (ante, No. 8), and is consequently separable; throwing off therefore all these special loci, we have a onefold locus which no longer comprises the points which correspond to curves passing through a node or a cusp of the given curve; the onefold locus, so divested of the special onefold factors, may be termed the "contact-locus" of the given curve. To each point of the contact-locus there corresponds a curve having with the given curve a two-pointic intersection, viz. this is either a proper contact, or it is a special contact, consisting in that the sought for curve has on the given curve a node or cusp, or (which is a higher speciality) in that the sought for curve is or contains as part of itself two or more coincident curves (ante, No. 3). To a point in general on the contact-locus there corresponds a curve having a proper contact with the given curve, save and except that to each point on any one of certain special loci on the contact-locus there corresponds a curve having some kind of special contact as above with the given curve. To fix the ideas, it may be mentioned that for the curves of the order r which touch a given curve of the order m and class n, the order of the contact-locus is $= n + (2r - 2)m$.

Article 387 (14)**.** If, then, the curve touch a given curve, the parametric point is situate on the contact-locus of that curve. If it touch a second given curve, the parametric point is in like manner situate on the contact-locus of the second given curve, that is, it is situate on the twofold locus which is the intersection of the two contact-loci; and the like in the case of any number of contacts each with a distinct given curve. But if the curve, instead of ordinary contacts with distinct given curves, has either a contact of the second, or third, or any higher order, or has two or more ordinary or other contacts with the same given curve, then if the total manifoldness be $= k$, the parametric point is situate on a k-fold locus, which is given as a singular locus of the proper kind on the onefold contact-locus; so that the theory of the contact-locus corresponding to the case of a single contact with a given curve, contains in itself the theory of any system whatever of ordinary or other contacts with the same given curve, viz. the last-mentioned general case depends on the discussion of the singular loci which lie on the contact-locus. And similarly, if the curve has any number of ordinary or other contacts with each of two or more given curves, we have here to consider the intersections of singular loci lying on the contact-loci which correspond to the several given curves respectively, or, what is the same thing, to the singular loci on the intersection of these contact-loci; that is, the theory depends on that of the contact-loci which belong to the given curves respectively.

Article 388 (15)**.** Suppose that the curve which has to satisfy given conditions is a line; the equation is $ax + by + cz = 0$, and the parameters (a, b, c) are to be taken as the coordinates of a point in a plane. Any onefold condition imposed upon the line establishes a onefold relation between the coordinates (a, b, c), and the parametric point is situate on a curve; a second onefold condition imposed on the line establishes a second onefold relation between the coordinates (a, b, c), and the parametric point is thus determined as a point of intersection of two ascertained curves. In particular if the condition imposed on the line is that it shall touch a given curve, the locus of the parametric point is a curve, the contact-locus; (this is in fact the ordinary theory of geometrical reciprocity, the locus in question being the reciprocal of the given curve; and the case of the twofold condition of a contact of the second order, or of two contacts, with the given curve, depends on the singular points of the contact-locus, or reciprocal of the given curve; in fact according as the line has a contact of the second order, or has two contacts with the given curve (that is, as it is an inflexion-tangent, or a double tangent of the given curve), the parametric point is a cusp or a node on its locus, the reciprocal curve; this is of course a fundamental notion in the theory of reciprocity, and it is only noticed here in order to show the bearing of the remark (ante, No. 14) upon the case now in hand where the curve considered is a line.

Article 389 (16)**.** If the curve which has to satisfy given conditions is a conic

$$(a, b, c, f, g, h \mid x, y, z)^2 = 0,$$

we have here six parameters (a, b, c, f, g, h), which are taken as the coordinates of a point in 5-dimensional space. It may be remarked that in this 5-dimensional space we have the onefold cubic locus $abc - af^2 - bg^2 - ch^2 + 2fgh = 0$, which is such that to any position of the parametric point upon it there corresponds not a proper conic but a line-pair; this may be called the discriminant-locus. We have also the threefold locus the relation of which is expressed by the six equations

$$bc - f^2 = 0, \quad ca - g^2 = 0, \quad ab - h^2 = 0, \quad gh - af = 0, \quad hf - bg = 0, \quad fg - ch = 0,$$

which is such that to any position of the parametric point thereon, there corresponds not a proper conic but a coincident line-pair. I call this the Bipoint-locus[(1)], and I notice that its order is $= 4$; in fact to find the order we must with the equations of the Bipoint combine two arbitrary linear relations,

$$(\alpha \,\ddot{\$}\, a, b, c, f, g, h) = 0,$$
$$(\alpha' \,\ddot{\$}\, a, b, c, f, g, h) = 0;$$

the equations of the locus are satisfied by

$$a : b : c : f : g : h = \alpha^2 : \beta^2 : \gamma^2 : \beta\gamma : \gamma\alpha : \alpha\beta$$

(where $\alpha : \beta : \gamma$ are arbitrary); and substituting these values in the linear relations, we have two quadric equations in (α, β, γ), giving four values of the set of ratios $(\alpha : \beta : \gamma)$; that is, the order is $= 4$, or the Bipoint is a threefold quadric locus.

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Article 390 (17). The discriminant-locus does not in general present itself except in questions where it is a condition that the conic shall have a node (reduce itself to a line-pair); thus for the conics which have a node and touch a given curve (m, n), or, what is the same thing, for the line-pairs which touch a given curve (m, n), the parametric point is here situate on a twofold locus, the intersection of the discriminant-locus with the contact-locus. It may be noticed that this twofold locus is of the order $3(n + 2m)$, but that it breaks up into a twofold locus of the order $3n$, which gives the proper solutions, viz. the nodal conics which touch the given curve properly, that is, one of the two lines of the conic touches the curve; and into a twice repeated twofold locus of the order $3m$ which gives the special solutions, viz. in these the nodal conic has with the given curve a special contact, consisting in that the node or intersection of the two lines lies on the given curve. By way of illustration see Annex No. 2. But the consideration of the Bipoint-locus is more frequently necessary.

Article 391 (18). Suppose that the conic satisfies the condition of touching a given curve; the parametric point is then situate on a onefold contact-locus $(a, b, c, f, g, h)^{\mu} = 0$ (to fix the ideas, if the given curve is of the order m and class n, then the order μ of the contact-locus is $= n+2m$). The contact-locus of any given curve whatever passes through the Bipoint-locus; in fact to each point of the Bipoint-locus there corresponds a coincident line-pair, that is, a conic which (of course in a special sense) touches the given curve whatever it be; and not only so, but inasmuch as we have a special locus of the same manifoldness as the general locus (ante, No. 8), it is consequently separable; throwing off therefore all these special loci, we have a onefold locus which no longer comprises the points which correspond to curves passing through a node or a cusp of the given curve; the onefold locus, so divested of the special onefold factors, may be termed the "contact-locus" of the given curve.

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Article 392 (19). The theory of the contact-locus is thus seen to be fundamental for the whole theory of the conditions which a curve may satisfy. The contact-locus of a given curve is the locus of the parametric point when the curve which has to satisfy the condition touches the given curve; and the theory of the singular loci on the contact-locus contains the theory of all the contacts of higher order, or of multiple contacts, with the same given curve. The theory thus presents itself in a form which is capable of extension to curves of any order and to any system of conditions.

Article 393 (20). The memoir contains also the discussion of several particular cases, which serve to illustrate the general theory. The case of the conic is treated in detail, and the contact-locus is determined for various kinds of given curves. The case of the cubic is also discussed, and the number of cubics which satisfy given conditions is determined in several cases. The general theory is thus verified by its application to particular cases, and its utility is established.

Article 394 (21). The theory of the curves which satisfy given conditions is thus brought into connexion with the general theory of loci in space of any number of dimensions. The parametric point, the conditions, the loci, the special solutions, and the proper solutions, are all elements of a general theory which is capable of application to a wide range of problems. The memoir thus forms a contribution to the science of geometry which is of permanent value, and which will serve as a foundation for future investigations.

Article 395 (22). The author expresses his hope that the theory developed in the present memoir may serve to advance the science of geometry, and to facilitate the solution of problems which have hitherto been regarded as of insuperable difficulty. The determination of the number of curves of given order which satisfy given conditions is a problem of fundamental importance in geometry, and the method of the present memoir furnishes a complete and general solution of this problem.

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Article 396 (23). The theory of the contact-locus has been applied in the memoir to the determination of the number of conics which satisfy various systems of conditions. The number of conics which touch five given curves is determined; the number of conics which pass through four points and touch a curve; the number of conics which pass through three points and touch two curves; and so on. The results are verified by comparison with known results, and the method is shown to be of general application.

Article 397 (24). The memoir also contains a discussion of the theory of the node and the cusp as conditions imposed upon a curve. A node is a twofold condition, and a cusp is a threefold condition; and the theory of these conditions is developed in connexion with the general theory of the present memoir. The number of curves of given order which have a given number of nodes and cusps is determined, and the results are verified by known formulae.

Article 398 (25). The theory of the inflexions and double tangents of a curve is also discussed in connexion with the Plücker equations. The number of the inflexions and double tangents of a curve of given order is determined by the Plücker equations, and these equations are applied to the curves which satisfy given conditions. The theory of the deficiency of a curve is also considered, and its connexion with the number of the conditions is pointed out.

Article 399 (26). The memoir concludes with a general survey of the theory which has been developed. The method of the parametric point and the loci in space of any number of dimensions is shown to be applicable to a wide range of problems; and the theory of the contact-locus, with its singular loci, is established as the foundation of the theory of contacts. The memoir thus forms a systematic development of the theory of the curves which satisfy given conditions, and it renders possible the solution of problems which have hitherto been regarded as of extreme difficulty.

Article 400 (27). The author expresses his indebtedness to the work of Chasles, Zeuthen, and others, who have treated particular cases of the general problem; and he hopes that the general theory which he has established may serve to unify and extend their results. The memoir thus forms a contribution to the science of geometry which is of permanent value, and which will serve as a foundation for future investigations.

Article 401 (28). In conclusion, the author remarks that the theory of the curves which satisfy given conditions is a branch of the general theory of loci in space of any number of dimensions. The problems which arise in this theory are of the utmost generality, and their solution requires the employment of the most powerful methods of modern geometry. The memoir is an attempt to establish these methods on a firm foundation, and to apply them to the solution of the most general problems of the theory.

Article 402 (29). The memoir thus concludes with the expression of the hope that the methods and results which have been developed may serve to advance the science of geometry, and to open up new fields of research in the theory of curves and their conditions. The theory is capable of extension in many directions, and it is the author's intention to pursue these extensions in future memoirs.

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